INTRAGEN.

DM Export Tool

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DM Export Tool

User Documentation — PowerShell 5.1

1. INTRODUCTION

The DM Export Tool extracts objects from One Identity Manager (OIM) transport packages and converts them into Deployment Manager (DM) compatible files. It supports three categories of export:

- Database objects (any OIM table row) — exported as DM XML or CSV+Template files.
- Processes (JobChains with Jobs, Events, Parameters) — exported as DM XML files.
- Scripts (DialogScript, ColumnTemplates, TableScripts, FormatScripts, CanSee/CanEdit scripts) — exported as VB.NET files.

This document covers the **PowerShell 5.1 version**, which runs on Windows PowerShell (`powershell.exe`) built into Windows. No additional PowerShell installation is required.

2. PREREQUISITES

Requirement	Details
Windows PowerShell 5.1	Built into Windows 10 / Server 2016+. Run `$PSVersionTable.PSVersion` to verify.
One Identity Manager	A running OIM instance with database access.
Deployment Manager DLL	`Intragen.Deployment.OneIdentity.dll` — included in `DeploymentManager-3.1.0/`.
NLog.dll	Typically found at `C:\Program Files\One Identity\One Identity Manager\NLog.dll`.
DM Project Config	A `projectConfig.xml` with a valid DataSource connection string.

3. FIRST-TIME SETUP

STEP 1: CONFIGURE CONFIG.JSON

You can edit `config.json` manually or use the **Configurator GUI** (Section 5). The file is located at `Powershell 5.1/config.json`.

```
{
  "DMConfigDir":         "C:\\path\\to\\Deployment\\Config\\Example",
  "OutPath":             "C:\\path\\to\\output",
  "LogPath":             "C:\\path\\to\\logs",
  "NLoggerDLL":          "C:\\path\\to\\NLog.dll",
  "DMDll":               "C:\\path\\to\\Intragen.Deployment.OneIdentity.dll",
  "DMPassword":          "",
  "IncludeEmptyValues": true,
  "PreviewXml":          false,
  "CSVMode":             false,
```

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```
  "ExcludedColumnsCSV": "C:\\path\\to\\Excluded.csv",
  "TableNameMapCSV":    "C:\\path\\to\\TemplateMap.csv",
  "ReportOnDenial":     true
}
```

Important: *The shipped config.json may contain outdated paths. Update all paths to match your environment before first use.*

Required fields: `DMConfigDir` must point to a folder containing `projectConfig.xml`; `DMDll` must point to the DM DLL file.

Password options: leave `DMPassword` empty to connect without a password; set a plain-text value; or set an encrypted `[E]...` value (see Section 9).

STEP 2: PREPARE TRANSPORT ZIP

You need an OIM transport ZIP file to process, exported from the OIM Designer or Administration Portal.

4. THE LAUNCHPAD

The Launchpad is the main hub for accessing all tools. It loads the DM module on startup and presents three tool cards.

HOW TO LAUNCH

```
powershell.exe -STA -File "Powershell 5.1\Launchpad_PS5.1 1.ps1"
```

STARTUP SEQUENCE

- Reads `config.json` for `DMDll` and `DMConfigDir`.
- Loads the DM module and connects via `Invoke-QDeploy`.

TOOL SELECTION

After startup, the Launchpad presents three tool cards. Click **Run** on any card to open that tool in a new `powershell.exe` window.

- **DM Exporter** — Launch the export wizard to select a ZIP file and run the export pipeline with optional CSV mode.
- **DM Change Explorer** — Browse and compare changes, inspect objects, and work with change labels.
- **DM Configurator** — Build and save `config.json` (paths, libraries, CSV settings, options).

5. THE CONFIGURATOR

The Configurator provides a GUI for building and saving `config.json` without manually editing the file.

HOW TO LAUNCH

- From the Launchpad: click **Run** on the "DM Configurator" card.
- Standalone: `powershell.exe -STA -File "Powershell 5.1\Configurator_PS5.ps1"`

GUI LAYOUT

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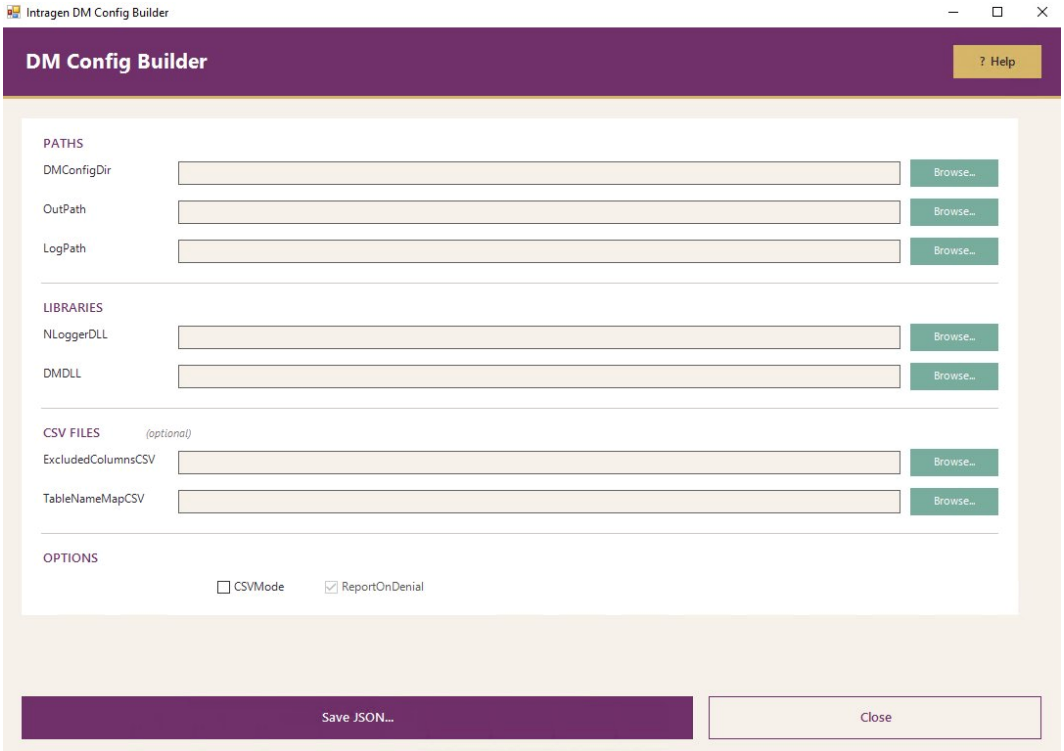

Figure 1. DM Config Builder — PATHS, LIBRARIES, CSV FILES, and OPTIONS sections.

CONFIGURATION SECTIONS

PATHS

- **DMConfigDir** — Folder containing the DM project config. Required.
- **OutPath** — Where exported files are saved. Required.
- **LogPath** — Where log files are written. Optional.

LIBRARIES

- **NLoggerDLL** — Path to `NLog.dll`.
- **DMDLL** — Path to `Intragen.Deployment.OneIdentity.dll`. Required.

CSV FILES (OPTIONAL)

- **ExcludedColumnsCSV** — Columns to exclude from exports.
- **TableNameMapCSV** — Maps table names to friendly display names.

OPTIONS

- **CSVMode** — Export DbObjects as CSV+Template pairs.
- **ReportOnDenial** — Always enabled.

Click **? Help** to toggle the field reference sidebar. Click **Save JSON...** to write `config.json`; overwrites are confirmed with a prompt.

6. THE EXPORTER (LAUNCHER)

The Exporter is the primary tool for processing transport ZIPs and running the full export pipeline.

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HOW TO LAUNCH

- From the Launchpad: click **Run** on the "DM Exporter" card.
- Standalone: `powershell.exe -STA -File "Powershell 5.1\Launcher_PS5.ps1"`

GUI LAYOUT

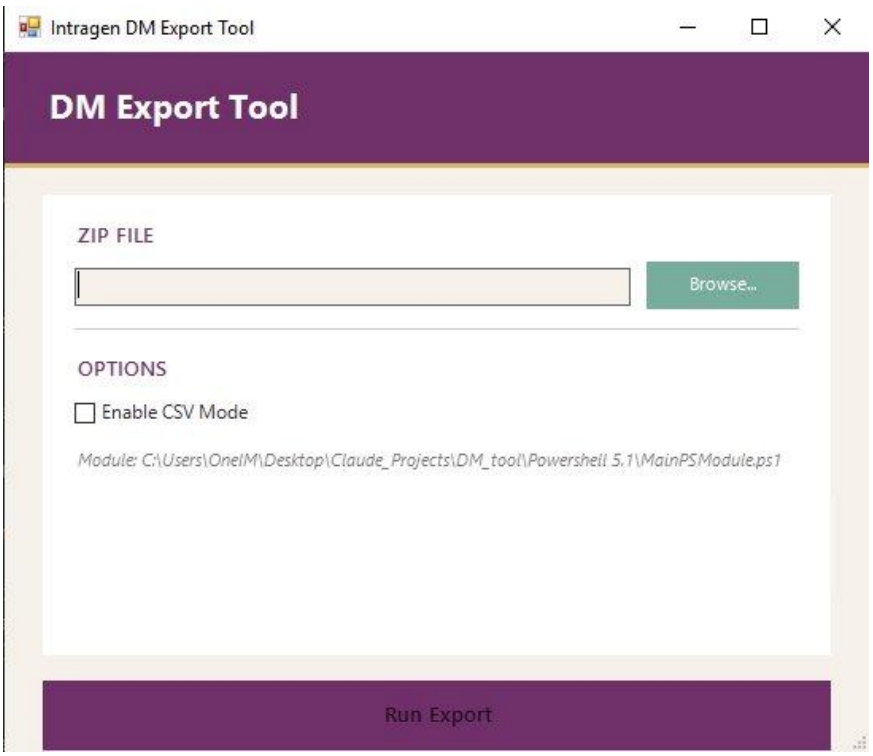

Figure 2. DM Export Tool — ZIP file selector, CSV Mode checkbox, and Run Export button.

USAGE STEPS

- Click **Browse...** to select your OIM transport ZIP file.
- (Optional) Check **Enable CSV Mode** for CSV+Template export.
- Click **Run Export**. A new console window opens showing progress.

PASSWORD HANDLING

The Launcher GUI does not include a password field. Passwords are configured through `config.json` (set `DMPassword`) or via the `-DMPassword` CLI switch for an interactive prompt (see Section 8).

7. THE CHANGE EXPLORER

The Change Explorer lets you browse transport ZIP contents visually without running a full export.

HOW TO LAUNCH

- From the Launchpad: click **Run** on the "DM Change Explorer" card.
- Standalone: `powershell.exe -STA -File "Powershell 5.1\ChangeExplorer_PS51.ps1"`

FEATURES

- Visual browser for all objects in a transport ZIP.

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- Side-by-side diff view comparing transport values against live database values.
- Object tree organised by table and change label.
- Object detail panel showing all column values.

8. COMMAND-LINE USAGE

You can run the export pipeline directly without any GUI by invoking the main module from a PowerShell prompt.

SYNTAX

```
powershell.exe -File "MainPsModule.ps1" -ZipPath <path> [options]
```

PARAMETERS

Parameter	Type	Required	Description
`-ZipPath`	string	**Yes**	Path to the OIM transport ZIP file.
`-DMConfigDir`	string	No	DM config directory (overrides config.json).
`-OutPath`	string	No	Output directory (overrides config.json).
`-LogPath`	string	No	Log path (overrides config.json).
`-DMDll`	string	No	DM DLL path (overrides config.json).
`-DMPassword`	switch	No	Prompts for password interactively via Read-Host.
`-IncludeEmptyValues`	switch	No	Include columns with empty values.
`-PreviewXml`	switch	No	Print generated XML to console.
`-CSVMode`	switch	No	Use CSV+Template export mode.

> **Note:** *CLI parameters always override values from config.json.*

EXAMPLES

BASIC EXPORT USING CONFIG.JSON DEFAULTS

```
powershell.exe -File MainPsModule.ps1 -ZipPath "C:\Transports\export.zip"
```

CSV MODE WITH XML PREVIEW

```
powershell.exe -File MainPsModule.ps1 -ZipPath "C:\Transports\export.zip" -
CSVMode -PreviewXml
```

INTERACTIVE PASSWORD PROMPT

```
powershell.exe -File MainPsModule.ps1 -ZipPath "C:\Transports\export.zip" -
DMPassword
```

9. PASSWORD ENCRYPTION

The tool supports encrypting passwords using Windows Data Protection API (DPAPI) to avoid storing plain-text credentials in `config.json`.

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ENCRYPTING A PASSWORD

Run the encryption utility:

```
powershell.exe -File "Encrypt-Password.ps1"
```

The utility prompts for the password and outputs a portable encrypted block:

```
Enter password: ********
Encrypted block (copy everything including [E] prefix):

[E]01000000d08c9ddf0115d1118c7a00c04fc297eb01000000...

Verification: decrypted successfully
```

USING THE ENCRYPTED PASSWORD

Paste the full encrypted string (including the `[E]` prefix) into `config.json`:

```
{
  "DMPassword": "[E]01000000d08c9ddf0115d1118c7a00c04fc297eb01000000..."
}
```

Important: *Encrypted passwords are tied to the Windows user account and machine that created them. Moving config.json to a different machine or user profile requires re-encryption.*

10. EXPORT MODES

NORMAL XML MODE (DEFAULT)

Produces a single DM Objects XML file containing all database objects. Output path: `DBObjects/000-DBObjects_NNN_TIMESTAMP.xml`.

CSV+TEMPLATE MODE

Produces separate XML template and CSV data file pairs per OIM table. Enable by checking **Enable CSV Mode** in the Launcher GUI, or by passing `-CSVMode` on the CLI.

```
DBObjects/
  Template_TableName.xml        ← XML schema with @placeholder@ tokens
  001-TableName.csv             ← Data rows
  Template_LinkingSteps.xml     ← PWODecisionStep cross-references (if applicable)
  NNN-LinkingSteps.csv
```

11. STALENESS DETECTION

The tool compares each object's modification timestamp in the transport against the live database before writing output files.

THE EXPORT/ABORT DIALOG

When a stale object is detected, a dialog shows the object identifier and the conflicting timestamps:

```
+=====================================================+
|  Stale Object Detected                              |
+=====================================================+
|                                                     |
|  Object: Person (UID_Person = abc-123)              |
|                                                     |
|  Transport Date:  2026-03-01 10:00:00               |
|  Database Date:   2026-03-04 14:30:00               |
|                                                     |
|  Transport User:  admin                             |
```

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```
|  Database User:   jsmith                          |
|                                                   |
|        [ Export ]               [ Abort ]         |
+===================================================+
```

- **Export** — Continue processing. The object is marked as Valid in the report.
- **Abort** — Stop writing output files. Remaining objects are still scanned for the report, but the output directory is deleted.

12. THE HTML VALIDATION REPORT

After processing, an HTML report is generated summarising the validity of each exported object. The report is written to:

```
$OutPath/$TransportName/$ChangeLabelName/ExportReport.html
```

The report contains Valid/Invalid status badges per object, modification timestamps from both the transport and the database, and the responsible user for each object.

13. OUTPUT FOLDER STRUCTURE

All exported artefacts are written under the configured output path, organised by transport name and change label:

```
$OutPath/
  └── TransportName/
      └── ChangeLabelName/
          ├── DBObjects/
          ├── Processes/
          ├── Templates/
          ├── Scripts/
          ├── TableScripts/
          ├── FormatScripts/
          ├── CanSeeScripts/
          ├── CanEditScripts/
          └── ExportReport.html
```

14. TROUBLESHOOTING

"THE CONFIG.JSON DOES NOT EXIST"

Ensure `config.json` exists in the same directory as the script you are running. Use the Configurator to create one.

".NET ASSEMBLY VERSION MISMATCH"

If errors reference `System.Runtime Version=8.0.0.0`, the DLL was built for .NET 8 which is unavailable in PS5.1. Use a DM DLL version compatible with .NET Framework 4.x.

"FAILED TO CONNECT TO OIM"

- Verify `DMConfigDir` points to a valid `projectConfig.xml`.
- Check the DataSource connection string.
- If using an encrypted password, ensure it was encrypted on this machine by this user.

"PASSWORD DECRYPTION FAILED"

Re-encrypt the password on the current machine:

```
powershell.exe -File Encrypt-Password.ps1
```

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CONSOLE WINDOW CLOSES IMMEDIATELY

Launch with `-NoExit` to keep the window open after an error:

```
powershell.exe -NoExit -File MainPsModule.ps1 -ZipPath "..."
```

LOG FILE LOCATION

Check: `$LogPath/transport_log.log` or `$OutPath/Logs/transport_log.log`.

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